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Application for a Grant

Identification This page will be made available to selection committee members and external assessors.						
Funding opportunity Insight Grants				Funding Stream B (\$125,001 to \$500,000)		
Joint or special initiative						
Application title EquiTransit: Developing knowledge and tools to support equitable transit planning						
Applicant family name Páez			Applicant given name Antonio		Initials H	
Org. code 1350611	Full name of applicant's organization and department McMaster University School of Earth, Environment & Society					
Org. code 1350611	Full name of administrative organization and department McMaster University Research Office for Administration, Development and Support (ROADS)					
Language of the Application English				Preferred Adjudication Committee 435-09		
Does your proposal require a multi/interdisciplinary evaluation?				Yes	☐	No ☒
Is this a research-creation project?				Yes	☐	No ☒
Does your proposal involve Indigenous Research as defined by SSHRC?				Yes	☐	No ☒
Does your proposal involve human beings as research subjects? If "Yes", consult the <i>Tri-Council Policy Statement: Ethical Conduct for Research Involving Humans</i> and submit your proposal to your organization's Research Ethics Board.				Yes	☒	No ☐
Will any phase of the proposed research take place outdoors?				Yes	☐	No ☒
A. Will any phase of the proposed research take place on federal lands in Canada, other than lands under the administration and control of the Commissioner of Yukon, the Northwest Territories or Nunavut, as interpreted in section 2 of the Impact Assessment Act (IAA)?				Yes	☐	No ☒
B. Will any phase of the proposed research take place in a country other than Canada?				Yes	☐	No ☒
C. Will the grant permit a designated project (listed in the Physical Activities Regulations) to be carried out in whole or in part?				Yes	☐	No ☒
D. Will any phase of the proposed research activities depend on a designated project (listed in the Physical Activities Regulations) being led and carried out by an organization other than SSHRC?				Yes	☐	No ☒
	Year 1	Year 2	Year 3	Year 4	Year 5	Total
Total funds requested from SSHRC	123,000	92,910	91,460	90,410	101,760	499,540



Family name, Given name

Páez, Antonio

Participants

List names of your team members (co-applicants and collaborators) who will take part in the intellectual direction of the research. Do not include assistants, students or consultants.

Role

Co-applicant ☐

Collaborator ☒

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Department/Division name

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Department/Division name

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Role

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Collaborator ☐

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Higgins

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Org. code

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Full organization name

University of Toronto

Department/Division name

Geography



Research Activity

The information provided in this section refers to your research proposal.

Keywords

List keywords that best describe your proposed research or research activity. Separate keywords with a semicolon.

transit; urban planning; geography; equity; justice; operations; social inclusion

Disciplines

- Indicate and rank up to 3 disciplines that best correspond to your activity.

Rank	Code	Discipline	If "Other", specify
1	61899	Other Geography	Transportation Geography
2	61412	Urban Planning	
3			

Areas of Research

Indicate and rank up to 3 areas of research related to your proposal.

Rank	Code	Area
1	370	Transportation
2	350	Social development and welfare
3	320	Politics and government

Temporal Periods

If applicable, indicate up to 2 historical periods covered by your proposal.

From	To
Year BC AD ☐ ☐ ☐ ☐	Year BC AD ☐ ☐ ☐ ☐



Family name, Given name

Páez, Antonio

Research Activity (cont'd)

Geographical Regions

If applicable, indicate and rank up to 3 geographical regions covered by or related to your proposal. Duplicate entries are not permitted.

Rank	Code	Region
1	1120	Central Canada
2		
3		

Countries or Regions

If applicable, indicate and rank up to 5 countries covered by or related to your proposal. Duplicate entries are not permitted.

Rank	Code	Country or Region	Prov./ State
1	1100	Canada	ON
2			
3			
4			
5			



Family name, Given name

Páez, Antonio

Response to Previous Critiques - maximum one page

Applicants may, if they wish, address criticisms and suggestions offered by adjudication committees and external assessors who have reviewed previous applications.



Family name, Given name

Páez, Antonio

Summary of Proposal

The summary of your research proposal should indicate clearly the problem or issue to be addressed, the potential contribution of the research both in terms of the advancement of knowledge and of the wider social benefit, etc.

Transit systems are a key component of urban public policy. These systems intersect with broader transportation and urban land use issues and are therefore of great interest and practical import. In general terms, transit systems generate accessibility to opportunities that contribute to the economic operation of urban regions. But this is not the only service that transit systems provide, since they use space more efficiently than low-capacity vehicles (mainly private automobiles), and thus help to reduce congestion and result in lower emissions per capita. Furthermore, transit is an obligated form of transport for many members of the public who may not be able to own or use a private vehicle and increasingly supports many urban residents who choose not to own a vehicle. As such, transit systems also help to reduce transportation-related social exclusion, and support democratic participation in many essential activities.

Despite their importance in public policy, transit services are often offered without explicit mention of the principles that underpin their provision. Implicitly, transit is often seen as an economic service that must remain fiscally responsible. But while fiscal responsibility is an important consideration, it is not by any means the only one. In reality, many of the benefits of transit systems are non-monetary, and a tendency to see them mainly as economic services leads to a narrow approach that can blunt their attractiveness. Particularly, there are numerous examples where this leads to major inequities in terms of the distribution of the benefits and the costs of public transportation, while ignoring its role as a facilitator of democratic participation in society.

Planning for transit systems that are equitable and promote ideals of transportation equity and justice is an urgent task, especially as the limits of highly individualized transportation systems become clear. EquiTransit proposes to re-envision transit from a broader, more just perspective, and aims to create the knowledge and tools needed to realize this vision. This project follows on the heels of recent research initiatives in Canada, particularly the SSHRC-funded project Mobilizing Justice, which have broken new ground and made considerable progress towards more inclusive, equitable, and just transportation systems in Canada. EquiTransit builds on these foundations with a more targeted focus on public transportation.

Accordingly, the objective of EquiTransit is to generate novel, theoretically-sound and analytics-based equity indicators useful to assess the performance of public transit systems in Canada. Theoretically, we will use a framework of justice that emphasizes the 5WH approach (why, what, who, where, when, how). To ensure that our indicators are pertinent to the question of "why equity?", we propose a Codesign approach, whereby the research team will work with transit operators and users in the explicit identification of the rationale and motivation for the development and use of indicators. We plan to use cutting-edge geospatial technologies to investigate the questions "who seeks equity, and where and when?". Further, we propose to create open-source tools, including equity-informed planning and assessment tools, and place them at the fingertips of stakeholders at the lowest possible cost. Finally, in the spirit of Codesign, the effectiveness of the tools will be assessed in consultation with stakeholders, so that refinements can be incorporated in successive releases of the tools. In this way, EquiTransit aims to produce actionable knowledge to support the development of more equitable transit in Canada, to the benefit of urban residents, democratic governance, and democratic participation in Canadian society.

Detailed Description (max. 6 pages)

Objectives: Generate novel theoretically-sound and analytics-based equity indicators to assess the performance of public transit systems in Canada. Investigate their reliability by working with planners and transit users in areas with equity gaps. Create open-source tools, including dashboards, to make these indicators available to stakeholders. Assess the effectiveness and accessibility of the tools in consultation with stakeholders and incorporate feedback in newer releases of the tools.

Rationale: Traditional public transport performance evaluations usually rely on operational efficiency (1–3), reliability (4), safety (5), and objective comfort, but often leave aside equity considerations related to both the spatial and temporal coverage of transit service (6,7). Consequently, they often fail to meet the needs of the most disadvantaged residents (8). The transit planning process primary focus is on service cost (9–11), operational features (12–14), and emissions (12,15,16). Furthermore, current standards (e.g., Transit Capacity and Quality of Service Manual (TCQSM)) and manuals (e.g., Federal Transit Administration (FTA)) completely overlook equity dimensions, yet equity-deserving groups are those who rely most on public transit (17–20). When service is insufficient (21,22), they face the greatest risk of social isolation due to transport-related factors, preventing them from participating in essential activities (8,23,24). This group, characterized as captive-users, faces greater limitations caused by disparities in transit level of service (LoS), both spatially and temporally.

Approaches: The project aims to produce actionable knowledge to support the development of more equitable transit in Canada. At one level, we will produce equity indicators grounded in sound theories of justice. We consider this aspect of the project the one of greatest academic interest. Further, to ensure that the indicators are of practical relevance, they will be developed in partnership with stakeholders. A key aspect of procedural justice is ensuring that all stakeholders participate in the planning process and have access to the information they need for decision-making. For this reason, we: 1) adopt a codesign framework to design equity indicators in partnership with stakeholders in Hamilton, Ontario; and 2) propose to develop open-source tools to mobilize this knowledge. General Transit Feeds Specifications (GTFS) are the standard for evaluation and planning of transit operations, and will be used to implement an equity analytical module to complement the usual cost-based evaluation of transit systems. To facilitate the exploration of indicators and the evaluation and planning with the equity evaluation module, we will develop open-source tools (EquiTransit) that will embed all elements in an accessible interface (a dashboard). The tool will be based on open data and will be able to use data from different regions. The aim of the tools is to allow for the examination of socioeconomic and spatial-temporal inequalities in public transportation services, based on geo-spatial technologies.

Anticipated Benefits: The impacts of inequitable access to public transit are thought to be large, yet spatial-temporal inequalities in transit provision remain little understood and of peripheral interest in transit planning. Clear and transparent tools will allow these considerations to be integrated into planning practice and public discourse. Codesigning the indicators and tools can empower policy makers to make bolder decisions with greater public support. Including equity indicators in transit performance assessments will help uncover inequalities, enabling planners and advocates to understand how disadvantaged populations in different locations and time periods are, or are not, served by public transportation. For example, visualizations showing the frequency of public transit service across areas of different income levels and at different times of day can help practitioners identify where and when low-income populations are underserved, highlighting where interventions are most needed. Performance metrics will be disaggregated by income, race, gender, and other relevant population characteristics, in addition to their traditional spatial and temporal dimensions.

Facilitation of knowledge transition: This project aims to develop EquiTransit for planners and community members to better identify who has the most and least amounts of different transportation services, but even more crucially, where those populations are located and when are the services needed. Using spatial and temporal visualizations, planners and other stakeholders will be better able to address issues in equity in transportation access. This would be particularly beneficial for transit agencies to better understand where service changes or additions should be made to increase transit equity.

Public Transportation and Equity: Traditional transit performance evaluations optimize service for an average user by focusing on efficiency, ridership, and reliability. These are system level, operational performance metrics that frame transit evaluation in a way that does not engage with social outcomes. Yet, transit is most heavily relied upon by individuals who have the fewest alternatives (i.e., captive users). Evidence from Canadian sources supports this concern. In 2021, 17.6% of racialized commuters in large urban centres took public transit to work, compared with 6.4% of non-racialized commuters, and 13.1% of immigrants relied on transit versus 4.7% of those born in Canada (27). By May 2025, roughly one in four Black (25.7%), Latin American (25.3%), and Filipino (24.4%) commuters used transit, and the average transit commute had reached 44.1 minutes, nearly twice that of a car commute, with most racialized groups facing systematically longer journeys (28). These patterns coincide with material deprivation. Allen & Farber (25) estimate that close to one million Canadians across the eight largest CMAs live in low-income households in areas of poor transit accessibility, placing them at risk of transport poverty. Women in lower-income households also undertake substantially more care-related travel that are poorly serviced by current network designs (30).

This makes the distribution of service to different people, at diverse places, and various times a question of justice (25). Distributive justice in transportation asks who benefits from the system and who carries its burdens, shifting the evaluative criterion from system performance to the accessibility opportunities delivered to different socioeconomic and demographic groups (26). This framework moves beyond horizontal equity, which treats all riders identically, to vertical equity, that is, actively directing better service, coverage, and frequency to the groups who rely on it most and face the greatest systemic barriers. Vertical equity represents a stricter idea of justice, as new benefits are unequally distributed in favor of under-privileged populations.

Existing planning tools rarely delve into these inequities. A recent analysis of Toronto's bus network shows that although two-thirds of TTC riders are equity-deserving, most live outside the agency's designated equity priority areas, leaving much of the population the framework is meant to serve largely invisible to it (20). Traditional performance metrics, and even most accessibility measures, also average across the day, masking systematic temporal under-service during evenings, nights, and weekends, when shift workers, who are disproportionately racialized, immigrant, and low-income, most depend on transit (17). EquiTransit will close these gaps with indicators and tools that bring distributive justice into transit planning across both space and time.

WORK PACKAGES

WP.1 Theoretically-grounded equity indicators: It is well established that the burdens and benefits of transportation are distributed unevenly across space and time, with some groups receiving greater benefits at the expense of others (27). To better understand how these benefits and burdens are distributed among a diverse population, a theoretical framework related to equity and justice is needed. In this proposal, we follow recent work of the PI with members of the SSHRC-funded project Mobilizing Justice team (29) that adapted the framework of Jaggard (30) to the needs of transportation concerns. According to this framework, the examination of equity, fairness, and justice is guided by six interrelated questions: *Why?*, *Who?*, *What?*, *Where?*, *When?*, and *How?*, or the 5WH framework.

Why? Justice is a broad moral and political concept that relates to: (1) the rights and entitlements that should be recognised and enforced; (2) the fairness of processes and procedures in decision-making and distribution (procedural justice); and (3) the appropriate distribution of benefits and burdens of social life in society. The last concept, also known as distributive justice, is the most studied form of justice in the transportation literature and aims to provide a fair redistribution to repair past harms (31). This is not a technical question, but a question of values, and therefore it is incumbent on the research team to involve stakeholders to avoid imposing perspectives that may be extraneous to the community the research is meant to benefit. CODESIGN (WP.2) is essential to this purpose.

The questions of *Who?*, *What?*, *Where?*, and *When?* in the 5WH framework all intersect with the spatial and temporal distribution of transit and the populations it serves. For this reason, they are better investigated from a geo-spatial perspective (WP.3). These questions are described next.

Who? Disadvantaged population groups, including those affected by low income, limited ability, and historical discrimination, have become a key focus of transportation research (31,32). This is due to the potential of historical discriminatory policies and other disadvantages to negatively impact their transport experiences. Additionally, these groups have often been excluded from the transportation planning process, resulting in their needs frequently being not recognized. Relevant groups include youth, adults, women, children, low-income individuals, zero-vehicle households, people with disabilities, single-parent households, refugees, and racialized populations (33–38).

What? In this research proposal, we argue that public transportation should be the focus of distributive justice. Public transportation is essential for promoting more sustainable cities due to its potential to reduce traffic congestion and carbon emissions and to improve air quality (39–41). In addition, public transportation increases people's accessibility, the main benefit of the transportation system (27), understood as the ease of reaching spatially distributed opportunities (42,43). In fact, public transportation provides access to opportunities for those who cannot - or prefer not to - drive, walk, or cycle. Access to opportunities and destinations is the most auspicious indicator for distributive justice (27), the most comprehensive indicator of the performance of land use and transport systems (44), and one of the main outcomes of spatial development (45) reduce the risk of social exclusion. Since transit and its output, accessibility, depend on space and time conditions, Scholars argue that this measure serves as a comprehensive and transparent indicator of transportation supply, with the additional advantage of being replicable across different jurisdictions and periods (45). Accessibility is a required condition for increasing people's freedom of choice and promoting equality of opportunities (45). An adequate level of access to opportunities, which can be offered by well-planned public transportation system, can increase the chances of attending a doctor's appointment, watching a movie at a cinema, engaging in a sporting activity in a park, or finding and keeping a job.

Where? By their very nature, interventions in the transportation sector are spatial interventions, and, due to the structuring effect of space (46), they inevitably generate inequality by benefitting those who live in or have access to the areas served by transportation infrastructure and services, while at the same time potentially harming others, either directly or indirectly (29). It is not uncommon for transportation policies to provide a sufficient or more than sufficient level of accessibility to the majority of people who own cars (47), while failing to ensure a basic level of accessibility for those who do not have access to a car, cannot use one, or do not wish to use one. To address this imbalance, improvements are needed in public transportation networks that provide users with a level of freedom comparable to that enjoyed by car users (48). The spatial dimension of analyses of transportation equity is often based on the concept of spatial equity, according to which cities should seek to distribute the benefits of transportation systems equally across space (31). Although transportation equity requires spatially

targeted policies and interventions (34), rather than broad approaches that encompass all of society, our perspective views equity primarily as an issue concerning people. Consequently, while we intend to identify areas with insufficient public transportation service and accessibility, these areas will be identified through measures and analyses focused on equity-deserving populations and facing insufficient access to public transportation, rather than solely on areas with low service. From this perspective, areas with a higher concentration of disadvantaged populations should take priority over wealthier areas, as disadvantaged groups tend to rely more on public transportation (28). Our work will be based on the City of Hamilton and its transit provider, Hamilton Street Railway (HSR).

When? To ensure justice in transportation, we evaluate the temporal aspects of transportation interventions. For instance, we investigate when a specific intervention takes place, the historical policy context in which it occurs, and for how long its benefits and burdens can be attributed to that intervention. In addition, the temporal dimension is central to our analysis because we are interested in investigating how equity-deserving populations may receive insufficient public transportation service, identifying not only the areas but also the times of day when this insufficiency is most pronounced.

The last question in our conceptual framework relates to How can measures be implemented to reduce inequities. The last work package of this project (WP.4) relates to the development of tools for equity-informed planning and evaluation. This part of the project intersects with WP.2 (CODESIGN) since these tools must implement concepts that are perceived as legitimate by all stakeholders.

How? Justice theories typically address sufficientarian and egalitarian moral concerns regarding how benefits and burdens should be distributed. While these two concepts are frequently applied in transportation equity research, they are often not explicitly articulated (22,49). The first approach, the sufficientarian perspective, focuses on why some groups or areas lack sufficient access to those goods and services. This approach builds on the idea of a ‘sufficient threshold’, that is, minimum levels of transport goods, services, and accessibility to essential activities that should be guaranteed to everyone to meet basic needs and lead to a meaningful life. In this idea, absolute priority should be given to improving the well-being of those whose well-being level falls below the sufficient threshold. Once that indicators have been selected and agreed upon by participants in the codesign part of the project, they will be implemented into equity-informed evaluation tools, as described in WP.4.

More generally, transportation analyses in practice often on standards to determine whether a situation is fair (29). Once operationalized, an equity standard defines the level of fairness to be adopted and the acceptable level of inequality, usually combining a concept of fairness with threshold criteria, thereby reflecting both vertical and horizontal approaches to equity. Vertical equity is typically operationalized through opportunity-based standards defined for specific population groups (e.g., minimum access to opportunities within a 10-minute bus ride for older adults), whereas horizontal equity is generally operationalized through uniform travel barrier thresholds applied to all potential users (e.g., a maximum acceptable distance to a public transit stop). Regarding the definition of equity standards, our approach is guided by the principles of sufficiency and egalitarianism. This means that the minimum level of accessibility to public transportation should be defined according to the needs of the most equity-deserving population, thereby reducing inequalities in opportunities.

WP.2 Codesign: In the 5WH framework, the question of WHY? constitutes the cornerstone of our approach to analyzing equity. This question, however, is not answerable in a strictly technical fashion. Data can help to support it, but to produce truly actionable knowledge public transit users and practitioners must co-create the equity indicators developed to assess the performance of public transport systems in Canadian cities. For this reason, CODESIGN is an essential component of the

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project, and will run throughout the life-course of the project, from the development of indicators, the choice of standards, and the testing of open-source tools.

Codesign is a form of community-based participatory action research; it is a collaborative approach that brings together researchers and stakeholders to address the needs of the community, enable social system change, and increase the impact of research. Stakeholders are placed at the centre of the design process to effectively solve problems and improve human experiences (50–52). In this project, the stakeholders will include the research team, public transit providers, and users.

Though the tools for this overarching proposed research project will be developed for Canada's 14 largest CMAs, the Codesign process will focus on the City of Hamilton, where McMaster is located and where the strongest community-based research already partnerships exist. Research collaborators working at the HSR will be involved as the public transit providers. Public transit users will be selected from Hamilton's 'code red' neighbourhoods; areas that experience the highest rates of poverty, unemployment, and chronic illness in the city (53).

The research will follow the 5-step human centered Codesign approach (54). Empathy is the first step. We will begin by understanding service users' experiences with public transport systems as well as their understandings of the standards the currently exist. This will be done through a series of Codesign focus groups. The second and third steps, to Define the key issues, and to Ideate, i.e., to brainstorm possible solutions, will take place in a second round of focus groups. This time, a series of separate Codesign focus groups will take place with users and with transit providers. PhD student 1 will lead these three sets of focus groups and present the results to the larger research team. The larger team will then complete step 4, which is to Prototype a variety of tangible solutions based on the focus group discussions so far. The final step is to Test and re-design the solution as needed. A final round of focus groups with users and providers in which they will present the prototyped solutions for feedback. These results will be presented to the research team again, with the possibility of another round of re-design and testing. The final outputs should include new tools that are not only theoretically and empirically sound, but that also respond to the needs of transit users and providers.

All Codesign focus group sessions will be supported by McMaster university's equity-based Co creation hub (eqCC). This hub unites researchers, students, service providers, and users to engage educate and innovate for systemic transformation (55). All focus groups will take place at McMaster University, and all participants will be compensated for their time with a \$50 gift cards per focus group they participate in. They will also be reimbursed for transportation (either reimbursed a bus fare or parking will be paid for).

WP.3 Geospatial data analytics: EquiTransit will be built entirely from open data and open-source software, ensuring transparency and reproducibility across jurisdictions. For each Census Metropolitan Area, we assemble four open inputs: sociodemographic data from the 2026 Canadian Census (via the {cancensus} interface to CensusMapper from von Bergman et al. (56), General Transit Feed Specification (GTFS) schedules published by transit agencies, OpenStreetMap road and pedestrian networks, and the Global Human Settlement Layer built-up surface. To enable consistent comparison, these inputs are harmonised onto a fine hexagonal grid; dasymetric interpolation, restricted to residential cells, apports census population and attributes to each hexagon. Using the {r5r} routing engine (57) we then compute different measures of transportation accessibility. This includes walking travel times between populated hexagons and transit stops to estimate the population effectively served by each stop. Larger network accessibility analyses will also be conducted to capture how well the transit system enables individuals to reach key destination types, such as places of work or daily necessities, around each study area within a reasonable travel time.

Critically, the travel time and accessibility analyses will be conducted using ten-minute transit service intervals, which is done to capture how level of service (e.g. service frequency) varies through the day and expose temporal gaps that daily averages conceal. GTFS Realtime data, which captures actual transit vehicle movements rather than static schedules, offers a particularly interesting avenue to better understand how service levels and accessibilities are provided. Building on initial schedule-based analyses, we will expand our lens to incorporate realtime data over sampled time windows as the project progresses, making use of new computational tools like the {gtfsrealtime} (58) package for R. From this, the transit accessibilities are disaggregated by income, race, gender, and immigration status, revealing inequities across both space and time. The complete workflow will be packaged as a reproducible analytical pipeline and an interactive dashboard, initially covering Canada's fourteen largest CMAs, with parameters, indicators, and presentation refined through stakeholder consultation.

WP.4 Equity-informed transit planning and evaluation: Transit planning is often planned on legacy routes and updated as population and urban geography expand. Then the service is packaged into a global universal standard: General Transit Feed Specification (GTFS). While the planning process itself is mathematically challenging, it is fundamentally developed under a cost-reduction framework, maximizing the utilization of resources, and balancing operation with political decision and budget constraints. The well-established four-step transit design model (e.g., routes, frequencies, fleet, and crew assignment) does not include socio-economic and demographic factors that capture the equity dimension users. We will develop two interventions enabling a novel implementation of equity transit planning.

For existing network, a GTFS-based equity analytical model will be conceived, developed and deployed. The model will utilize equity indicator, geospatial data, and findings from the CODESIGN, and integrate them with GTFS files. Furthermore, Equity KPIs will be embedded in a GTFS-analytical model, enabling transit agencies, policymakers, and practitioners to evaluate the equity gaps in existing design. For new service design, the equity indicators will be integrated in the four-step transit model, offering a novel equity-based transit planning framework transferable to any context. Equity-indicators will be integrated in all four steps of the planning process. For example, frequency estimation is based on passenger flow/demand, wait-time, and fleet size (e.g., available resources). Although passenger flow/demand is an indirect indicator of equity, it remains ineffective of differentiating service provision for equity-deserving groups compared to other users.

Table T1. Project timeline with Work Packages and Tasks.

Work Packages	Tasks	Year 1	Year 2	Year 3	Year 4	Year 5
WP.1 Theoretically-grounded equity indicators	Overall direction and transversal communication between work packages (Why? What? Who? Where? When? How?)	Antonio Pérez & Juanita De Barros				
		PD 1				PD 2
WP.2 Codesign	Equity indicators (Why?)	Dr. Léa Ravensbergen				
	EquiTransit tools (How?)	PHD 1			PHD 1	PD 2
WP.3 Geospatial data analytics	Equity indicators (What? Who? Where? When?)	Dr. Christopher Higgins				
	EquiTransit tools (How?)	PD 1	PHD 2			PHD2
WP.4 Equity-informed transit planning and evaluation	Development of base tools	Dr. Moataz Mohamed				
	Integration of equity indicators (How?)	M 1			M 2	

Personnel (Key). PD1, 2: Post-Doctoral Fellows; PHD 1, 2: Doctoral Students; M1, 2: Master's Students

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Knowledge Mobilization Plan (max. 1 page)

Appropriate Knowledge Mobilization (KM) strategies will be used to reach key audiences: 1) members of the public; 2) transportation planners and practitioners; and 3) academic and scientific peers. The creation of transparent, research-based tools will improve planning processes and outcomes, and support fluid participation in planning processes by a variety of stakeholders.

Public and practitioners (Years 1-5): A key aspect of the proposed work is the use of a codesign framework. In codesign there is a continuous and multilateral flow of knowledge during every step of the project. In addition to the involvement of members of the public and practitioners in this endeavor, we will expand the diffusion of this research by writing general reports in layperson language to facilitate knowledge transfer. These reports will take the form of policy briefs, user's manuals, and sample case reports related to the release of the tools. These will be posted publicly via a website that will be hosted on GitHub Pages. Members of the research team have experience developing and releasing available materials for public use with this medium (59,60). Communication with the public and practitioners will begin immediately through the codesign framework.

Academic (Years 2-5): This research proposal includes 4 research students (2 Master's and 2 PhDs), two post-doctoral researchers, five professors, and one international collaborator. The goal is for each Master's student to lead at least one paper for a national conference and then publish in an international peer-reviewed journal. PhD students will aim to publish three peer-reviewed journal articles. The post-doctoral fellows will lead one research paper each. Thus, 10 papers would be led by trainees. This project primarily has implications for planning in urban regions. Accordingly, target journals for this research include *Transportation Research A: Policy and Practice*; *Public Transportation*; *Transport Policy*; *Journal of Transport Geography*; and *Transportation*. Pre-print versions of papers will be published in their GitHub repositories. Conferences will target audiences with both researchers and practitioners. Further knowledge diffusion will be achieved via workshops in relevant conferences. Communication with academic audiences will begin in year 2 of the project when initial results can be reported.

Open-Science Statement: The Tri-Council system in Canada (i.e., CIHR, NSERC, SSHRC) is currently phasing in an Open Access policy for the research it funds. Open Access of publications and data are only one facet of open science, since encouraging quality of work throughout the research process requires an effort to share the data-collection process, methodology, and analysis as well (66). Open science can encourage and increase engagement from members of the public regarding ongoing research and any relevant issues that can emerge (61). In this respect, members of the research team have already a proven record of creating open, reproducible science, not only by publishing preprints publicly and sharing data, but by sharing every aspect of research, from the stage of data preparation and up to reproducible computational notebooks. Examples of data sharing by members of the research team include the creation of several open-data products (62) such as ActiveCA package (63) and TTS2016R package (59). An example of equity-based research in transportation that illustrates the mode of sharing pre-prints along with reproducible notebooks is Desjardins et al. (64). Materials such as this, shared publicly on websites like GitHub, allow for thorough verification of the research, to ensure it does not contain major errors that could impact the validity of the results (65). Open-science practices like this also speed up the process of discovery, by making it easy for other researchers to access the base work and modify it to accomplish whatever is required for their own specific work. The members of the research team are experienced open-science researchers and are fully committed to following Tri-Agency policy on Open Access, and more broadly to continue producing high-quality research that is open, transparent, and reproducible.



Family name, Given name

Páez, Antonio

Expected Outcomes

Elaborate on the potential benefits and/or outcomes of your proposed research and/or related activities.

Scholarly Benefits

Indicate and rank up to 3 scholarly benefits relevant to your proposal.

Rank	Benefit	If "Other", specify
1	Enhanced theory	
2	Enhanced research methods	
3	Student training/skill development	

Social Benefits

Indicate and rank up to 3 social benefits relevant to your proposal.

Rank	Benefit	If "Other", specify
1	Social outcomes	
2	Enhanced policy	
3	Enriched public discourse	

Audiences

Indicate and rank up to 5 potential target audiences relevant to your proposal.

Rank	Audience	If "Other", specify
1	Academic sector/peers, including scholarly associations	
2	Municipal government	
3	General public	
4	Practitioners/professional associations	
5	International audiences	



Family name, Given name

Páez, Antonio

Expected Outcomes Summary

Describe the potential benefits/outcomes (e.g., evolution, effects, potential learning, implications) that could emerge from the proposed research and/or other partnership activities.

The project assembles a group of researchers with a proven record of collaboration, and it fosters new collaborations between several research professors. This includes new collaborative opportunities between Ravensbergen and Higgins, Mohamed and Pereira, Ravensbergen and Pereira, and all transportation scholars collaborating with De Barros, an expert in human rights and justice. The team takes previous collaborations and expands them in novel ways that will benefit the quality of the scholarship produced. Moreover, the codesign framework proposed builds on existing collaborations (e.g., between Co-PI Ravensbergen and Mohammed and Hamilton's transit agency) and fosters new connections with community stakeholders. The project aspires to empower users of transit to have a more fluid participation in decision-making processes that affect them, and to help providers of a key public service to better incorporate user knowledge. In this way, the codesign framework opens avenues for knowledge exchange between providers and users of transit and thus contributes to enhance the process and practice of effective governance.

Through this project, four graduate students and two post-doctoral researchers will be trained and gain various new skills, to the betterment of Canada's transportation and research expertise. Other outcomes of the project are of three kinds: communication, research materials, and tools. The project will produce ten peer-reviewed papers (1 article by each Masters student, 3 articles per Doctoral student, and 1 article by each post-doctoral researcher). In addition, there will be communications for lay audiences, in the form of case studies, policy briefs, and user's manuals. New data produced during the research will be shared in the form of open data products, which will be made available via web repositories in addition to registering them for digital object identification (doi) using Zenodo. Other research materials, such as research documents and computational notebooks, will be shared using public web repositories. The project aims to push the boundaries of current practices in open science by expanding on the current expertise of members of the research team. The project will also produce open-source tools (e.g., dashboards) to help stakeholders assess the equity implications of various transit provision scenarios. More generally, knowledge transfer will prepare the next generation of transportation equity champions.

With better knowledge, better choices will be facilitated for planners and policymakers at transit agencies and users of public transportation. Rather than basing decisions on relatively narrow fiscal considerations, transparent and publicly available tools will contribute to advance more equitable public transportation planning. As a matter of procedural justice in transportation, planning must consider all segments of society, and for participation to be meaningful relevant information must be accessible. The tools proposed contribute to fill an information gap about levels of service by clarifying in a more holistic way the distribution of transit's costs and benefits. The development of visual tools based on research that accounts for relevant perspectives of stakeholders will enhance public discourse, public policy, and planning. Generating knowledge about the equity implications of transit operations will help steer conversations from a predominantly economic perspective to a fuller accounting of transit's numerous non-monetary benefits. Generating the tools to implement this knowledge will help inform policy and practice related to urban planning, transport, social inclusion, and community development by translating abstract research into trustworthy, locally relevant, and actionable information.

Research team, previous output and student training (max. 4 pages)

A. DESCRIPTION OF RESEARCH TEAM

The theme of public transportation and equity is multifaceted and requires an appropriate mix of skills and expertise. In response, this project gathers a group of researchers with complementary skills and perspectives. The research team is composed of one principal investigator, four co-investigators and one international collaborator, and includes scholars in the social sciences (Páez, Higgins, Ravensbergen, and Pereira), humanities (De Barros), and engineering (Páez and Mohamed). The principal investigator (Páez) and two of the co-investigators (Higgins and Ravensbergen) are alumni of the SSHRC-funded project Mobilizing Justice. One of the co-investigators (De Barros) is Director of McMaster Centre for Human Rights and Restorative Justice, and another (Mohamed) is Director of McMaster Institute for Transportation and Logistics. The profiles of the members of the research team, along with their expertise and roles in the project are as follows.

Juanita DE BARROS (co-applicant, relative contribution to project 15%, 15% of personal research time) is Professor of History at McMaster University and the director of the Centre for Human Rights and Restorative Justice. She is an internationally-recognized historian whose work explores the intersection of race, gender, and class in the construction of public policy, including in colonized urban spaces. This domain of knowledge is key to the project, as it combines social and government policy in ways that affect racialized and genderized people in urban spaces. De Barros expertise in discourse analysis and in the field of human rights is central to the aims of this project, where a clear understanding of stakeholders about the grounds for justice in public services is essential. For this project, she will lend her expertise and vision through the co-supervision of trainees.

Christopher D. HIGGINS (co-applicant, relative contribution to project 15%, 15% of personal research time) is Associate Professor at the University of Toronto Scarborough in the Department of Human Geography and in the Department of Geography and Planning at the University of Toronto. He is among the world's top 2% of researchers in the Elsevier-Stanford rankings and received the 2025 University of Toronto Scarborough's New Scholar Award for exceptional research achievement and impact among early career faculty. He has experience working with public sector partners such as the City of Toronto, City of Hamilton, and Ontario Ministry of Transportation. His research uses geospatial analysis and accessibility measurement to examine how transportation systems shape access to housing, employment, and essential services, with particular attention to social equity. His work has been supported by SSHRC and NSERC, most recently through an NSERC Alliance Grant supporting the University of Toronto's tri-campus Mobility Network Consortium. For EquiTransit, he will lead the development of geospatial analytics, accessibility indicators, reproducible data pipelines, and interactive analytical tools.

Moataz MOHAMED (co-applicant, relative contribution to project 15%, 15% of personal research time) is associate Professor of Smart Systems and Transportation at McMaster University and Director of the McMaster Institute for Transportation and Logistics (MITL). He is the 2025 recipient of the Joseph Ip Distinguished Engineering Fellowship and is featured Elsevier-Stanford list of the world's top 2% of scientists. Mohamad is an experienced researcher, with more than 90 peer-reviewed articles and over \$6 million in secured research funding to his credit. He has trained 57 highly qualified personnel, including 11 PhD students, 12 master's students, and 7 postdoctoral fellows. Since 2018, he has been principal investigator on a City of Hamilton-funded program to reenvision the city's transit operator, Hamilton Street Railway (HSR). He is the principal developer of the AI-Transit Orchestrator, an \$1.4M NRCan-funded fleet-planning toolkit co-developed with Halifax Transit, Calgary Transit, Niagara Region Transit, and ETUQ. On this proposal, he will lead the team's work to integrate equity

considerations in transit planning operations, translating multi-agency networks into a GTFS-based equity analytical model and equity-informed planning framework.

Antonio PÁEZ (principal applicant, relative contribution to project 30%, 80% of personal research time) is Professor of Transport Geography at McMaster University. He is a seasoned researcher who led the multi-institutional team that won the 2010 Meredith F. Burrill Award of the American Association of Geographers for exceptional work at the intersection of policy and basic research. He is also the recipient of the 2024 Edward L. Ullman Award of the American Association of Geographers for lifelong contributions to transportation geography. Páez has published widely, including over 170 peer-reviewed publications, three edited books, and a textbook. Páez is among the top 20 transportation researchers in the world whose careers started after 1999 according to Elsevier-Stanford's ranking, and is best known for his social science theory-informed quantitative research. This includes work on *normative accessibility* and *relative accessibility*, which aimed at implementing equity ideals in accessibility analysis. He is also well-versed in theoretical and qualitative approaches. He will direct the project and will participate in research and supervision of graduate students across the project.

Rafael H.M. PEREIRA (collaborator, relative contribution to project 10%, 5% of personal research time) is Director of Data Science at IPEA, Brazil's federal economics research institute. His contributions to transportation justice are largely responsible for shaping the field (27,32,66,67), while his contributions to numerous publicly available *R* and *Python* packages have been hugely influential in changing the practice of transportation research and planning. Open-source software such as {r5r} (57) and {r5py} (68) (which collectively have over a million downloads) are now routinely used by community planners, geographic researchers, and other members of the public for the purpose of simulating routing by different transportation modes. He has also worked on several packages for working with general transit feed specification data such as {gtfstools} (68). His role will be to provide advice on the theory and practice of transportation justice, and he will also lend his considerable expertise to the development of open-source tools.

Léa RAVENSBERGEN (co-applicant, relative contribution to project 15%, 15% of personal research time) is an early career researcher of equity in sustainable urban mobility. She is a mixed-methods researcher who has led multiple qualitative research projects focused on transportation users' experiences, including two that utilized a Codesign framework. In 2025, she led a series of Codesign focus groups with older adults on how best to design mobility reporting to encourage active lifestyles (currently under review in *Journal of Aging and Physical Activity*). In 2026, she worked with the Hamilton's transit operator Hamilton Street Railway (HSR) to Codesign how best to deliver paratransit as the City of Hamilton moves toward integrated transit. Ravensbergen will lead the Codesign component of the project.

B. DESCRIPTION OF PREVIOUS AND ONGOING RESULTS

Juanita DE BARROS's main field of research is into human rights and justice, with a focus on the political and social factors that shaped policies dealing with the intersection of health and the law in the past, their impact on the lived experience of women and children especially, and the longer-term impact of these policies. Her research projects include a sole-authored monograph on gender and reproductive health in the Caribbean (75), a forthcoming co-written article on urban vagrancy laws in late nineteenth-century Guyana, and a multidisciplinary co-edited essay collection (with McGill-Queens University Press). This latter project addresses child protection policies and discourse from the nineteenth to the early twenty-first centuries and the varied ways that these policies served adult institutional and political interests, often resulting in the forced labour and incarceration of children. This project is De Barros first foray into transportation-related research, and we believe that her historian's eye, in addition to her

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expertise in human rights and justice, will be an asset to the proposed work. Historical perspectives on transportation remain rare despite their value (76,77) and collaboration between historians and transportation scholars is a unique aspect of the project.

Christopher D. HIGGINS has done research with conceptual, methodological, and applied outputs that are relevant to the proposed work. This includes the development of place-based/person-based (69), as well as competitive accessibility measures (43,70) that improve our understanding of who can reach destinations using the transportation network. His applied studies have incorporated equity into route-level transit performance measurement and examined disparities affecting the access of low-income residents, and essential workers to health and social services (38,71,72). Complementary research has assessed relationships among transit accessibility, residential location, and housing prices within individual metropolitan regions and across Canada (73). His ongoing research extends these approaches to the joint analysis of equity in accessibility and housing affordability. Further, Higgins brings a commitment to open, replicable, and reproducible research as evidenced through the development of the {pynogrid} package (74) and his work on a forthcoming open access book “Advanced Accessibility Analysis with R”. These skills mean that EquiTransit is well-grounded for developing theoretically-informed equity indicators, for conducting comparable analyses across Canadian metropolitan areas, and for communicating complex patterns through accessible analytical tools.

Moataz MOHAMED is an expert in developing and evaluating indicators of transit service performance and quality from both operator and rider perspectives (78–80). He has led qualitative and mixed-methods investigations into the needs of current and potential riders of transit, including grounded-theory studies of the barriers travelers face (81,82). As well, he has studied the satisfaction of transit users among population with distinct mobility needs (83). Mohamed is an expert in travel behaviour and preference heterogeneity across rider personas and generational cohorts (84–86), and in the development of accessibility and spatial-analysis measures for transportation (70,87,88). He has direct experience translating research into open, web-based decision-support tools for public agencies and practitioners, including work applying open-source data and machine learning to inform transportation decision-making (89). He has developed public-facing tools for the Ontario Ministry of Transportation. Currently, Mohamed is Director of the McMaster Institute for Transportation & Logistics and PI/Co-PI on a portfolio of NSERC-, NRCan-, and municipally funded projects on electric and zero-emission transit system planning, resilience, and equitable transit service design.

Antonio PÁEZ (Principal Investigator) has extensive experience with the study of the transportation challenges faced by transport-poor populations, including older adults (90–93), single parents (94,95), and children (96–98). His longstanding interest in the implications of social science theory and policy on methodology is illustrated by his research on normative accessibility (99) and relative accessibility (100), work initially motivated by the policy needs of Human Resources and Social Development Canada (an agency of Canada’s federal government). His interests now encompass broader issues of equity and justice in transportation (64,71,101). As a member of the SSHRC-funded project Mobilizing Justice, his research led to novel tools for the analysis of accessibility (43,70,102), clearer theoretical foundations for developing standards of equity (45), and the role of human dignity in transportation (103). Páez is an expert in the development and application of spatial analysis tools with a focus on transportation (87,94,104) and the development of web-based tools for public use (105,106), and he is also fluent in the use of qualitative methods (82,107). His breadth of knowledge and expertise is an asset for this project, in his role as PI and transversal conduit between the various work packages.

Léa RAVENSBERGEN has researched the transit experiences of equity-seeking groups (83,108,109) and the systemic barriers transit agencies face in planning just systems (108,110,111). Her expertise is as a qualitative methods researcher (109,112–114) including mobile ethnography (115,116). She has conducted Codesign research in which older adults codeveloped a mobility reporting tool (117)

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and has expertise in community-based research (118). Her work includes collaborations with all levels of governments in various provinces, transit agencies across the country, and non-for-profit organizations. She is part of a pan-Canadian team that was granted a \$.16 million NSERC Alliance grant to research data-driven tools for planning and operating gender-inclusive public transit systems in Canada. In this role, she is working in partnership with 14 Canadian transit agencies to better integrate and implement gender-sensitive data collection practices. Since 2023 she has worked with the Hamilton Street Railway (HSR), the public transit agency of the City of Hamilton (108,119,120). She is well positioned to lead to the Codesign element of this research, and has the relationships required to ensure a successful process.

C. DESCRIPTION OF TRAINING STRATEGIES

Our strategy for training highly qualified personnel is based on extensive involvement of the students, well-defined responsibilities for the integration of graduate students into all aspects of the project, and close supervision by a team of interdisciplinary supervisors (see Table MT.1).

Table MT.1. Supervisory Matrix

			Year	Main Supervisor	Supervisory Committee
WP.1 Theoretically-grounded equity indicators	Post-Doctoral Fellow 1	PD1	1	Páez	De Barros, Pereira
	Post-Doctoral Fellow 2	PD2	5	Páez	De Barros, Pereira
WP.2 Codesign	Ph.D. Student 1	PhD2	1 - 4	Ravensbergen	Páez, De Barros
WP.3 Geospatial data analytics	Ph.D. Student 2	PhD1	2 - 5	Higgins	Páez, Pereira
WP.4 Equity-informed transit planning and evaluation	Master Student 1	M1	1 - 2	Mohamed	Páez, Higgins
	Master Student 2	M2	3 - 4	Mohamed	Páez, Higgins

PD1 and PD2 will work with Páez and De Barros across work packages. PD1 will be part of WP.1 and will interact mainly with the geospatial data analytics package (WP.3) on the data analytics needed for the codesign work package (WP.2) to communicate with stakeholders. In the latter part of the project, PD2 will liaise primarily with the codesign package (WP.2) to assess the usability and performance (user experience) of EquiTransit tools. PhD and Master's students will maintain a tighter focus. PhD1 will be on the codesign package (WP.2) and PhD2 will focus on geospatial technologies and the development of open-source tools (WP.3). M1 and M2 will participate in the development of equity-based assessment and evaluation, beginning with a base tool that will incorporate equity-informed indicators after they have been validated as part of the codesign package (see **Table T1**). Through the project, all trainees will develop a variety of soft and transferable skills (see **Table MT.2**) through a combination of close supervision and regular discussion, and enrolment in appropriate graduate courses [e.g., reproducible research; (121)].

Table MT.2. Skills Matrix

	Year	KS	QL	QT	PP	DL	PC	DM	RE	CCE	TW	IIC	TM	BM
PD 1	1													
PD 2	5													
PhD 1	1 - 4													
M 1	1 - 2													
M 2	4 - 5													
PhD 2	2 - 5													

Skills (Key): **KS:** Knowledge Synthesis; **QL:** Qualitative research methods; **QT:** Quantitative research methods; **PP:** Publication and presentations (communication skills); **DL:** Digital literacy; **PC:** Programming and coding; **DM:** Data management and analysis; **RE:** research ethics; **CCE:** Consultation and community engagement; **TW:** Teamwork; **IIC:** International and intersectoral conference; **TM:** Time management; **BM:** Budget management.



Family name, Given name

Páez, Antonio

Funds Requested from SSHRC

For each budget year, estimate as accurately as possible the research costs that you are asking SSHRC to fund through a grant. For each Personnel costs category, enter the number of individuals to be hired and specify the total amount required. For each of the other categories, enter the total amount required.

	Year 1		Year 2		Year 3		Year 4		Year 5	
Personnel costs	No.	Amount	No.	Amount	No.	Amount	No.	Amount	No.	Amount
Student salaries and benefits/Stipends										
Undergraduate										
Masters	1	25,000	1	25,000	1	25,000	1	25,000	0	0
Doctorate	1	28,000	2	56,000	2	56,000	2	56,000	1	28,000
Non-student salaries and benefits/Stipends										
Postdoctoral	1	70,000	0	0	0	0	0	0	1	70,000
Other										
Travel and subsistence costs										
		Year 1	Year 2		Year 3		Year 4		Year 5	
Applicant/Team member(s)										
Canadian travel										
Foreign travel										
Students										
Canadian travel		0		2,400		2,400		2,400		800
Foreign travel		0		3,900		3,900		3,900		1,300
Other expenses										
Professional/Technical services										
Supplies										
Non-disposable equipment										
Computer hardware		0		3,000		3,000		500		500
Other										
Other expenses (specify)										
Codesign materials/support		0		2,610		1,160		2,610		1,160
Total		123,000		92,910		91,460		90,410		101,760

Budget Justification (max. 2 pages)

The budget requested from SSHRC is \$499,540.

Personnel costs (\$464,000)

The largest portion of the costs are allocated towards the students (master's and PhD) and post-doctoral research fellows. Stipends for master's students are \$25,000 per year (years 1-2; 3-4), stipends of PhD students are \$28,000 per year (years 1-4; 2-5), and the Post-Doctoral Fellows will be paid \$70,000 per year (years 1 & 5). Over the duration of this 5-year project, this will account for 92.8% of the funds requested. A timeline of when master's students, PhD students, and post-doctoral fellows will be working on the project, and the tasks allocated to them, can be seen in **Table T1**.

Budget Item	Funds Requested
Student stipends	
Master's Student Stipends (4 student-years x \$25,000)	\$100,000
PhD Student Stipends (8 student-years x \$28,000)	\$224,000
Non-student salaries	
Post-Doctoral Salaries (2 person-years x \$70,000)	\$140,000
Total	\$464,000

Travel costs (\$21,000)

Beginning in year 2 of the project, each student will receive \$800 per year for Canadian travel to attend domestic conferences and meeting such as the Canadian Associations of Geographers Annual Conference or transportation-oriented conferences such as the Canadian Transportation Research Forum. This translates to \$2,400 in years 2-4 of the project, and \$800 in year 5, when the only student will be PhD2. Additionally, starting on year 2 of the project, each student will receive \$1,200 for international travel to conferences such as the Annual Meeting of the Transportation Research Board or NECTAR. This translates to \$3,900 per year in years 2 to 4 of the project and \$1,300 in year 5, when only PhD2 will be in the project. Regardless of year of study, students will be encouraged to attend the conferences as well as present in their upper years of study, as it is crucial for not only showcasing the research project, but also learning from other experts in the fields of transportation, geography, and planning. Hence, 4.2% of the total project cost will be allocated to travel costs.

Budget Item	Funds Requested
Travel and subsistence	
Students: Canadian travel (10 student-years x \$800)	\$8,000
Students: Foreign travel (10 student-years x \$1,300)	\$13,000
Total	\$21,000

Computer hardware (\$7,500)

The quantitative research segments of this project, namely the geospatial analysis, the development of the equity-informed evaluation tool, and the development of open-source packages, will require specialized equipment. This translates into computers with sufficient processors, RAM, and storage to be able to complete various computing processes and generating data. Members of the research team have some computers in their equipment inventory, and these will be used for this project. Funds are requested to upgrade or replace ageing equipment on years 2 and 3 of the project. We are budgeting \$3,500 in year 2 of the project, and \$3,000 in year 3 of the project to this end. In addition, we are

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requesting \$500 in each of years 4 and 5. These funds will be allocated to hardware updates, computer repairs, and the acquisition of various peripheral equipment as needed. Together with computer equipment already owned by members of the research team, the aim is to maintain four fully equipped workstations for the duration of the project, with one workstation assigned to each of the work packages.

Budget Item	Funds Requested
Non-disposable equipment	
Acquire computer equipment/replace ageing equipment (Year 2)	\$3,000
Acquire computer equipment/replace ageing equipment (Year 3)	\$3,000
Repair/peripherals (year 4)	\$500
Repair/peripherals (year 5)	\$500
Total	\$7,000

Other expenses: Codesign (\$7,540)

The qualitative research component of the project is the CODESIGN, with an estimated budget of \$6,000 to provide incentives for participants across different focus groups (based on factors such as socioeconomic status and/or neighbourhood). Each codesign focus group participant will receive a \$50 gift card and their transport to and from McMaster reimbursed ($\$4.00 \times 2$ (return bus fare) = \$8.00). All co-design focus group materials (e.g., printing, Bristol boards, etc.) will be provided by Ravensbergen's lab in kind. There will be two codesign phases: development of indicators and development of tools.

Budget Item	Funds Requested
Other: Codesign (Development of indicators)	
Step 1 (Empathy) Four focus groups with 5 stakeholders (Year 2) (4 x 5 x \$58)	\$1,160
Steps 2 and 3 (Define and Ideate) Three focus groups with 5 transit users and 2 focus groups with 5 transit providers (Year 2) (3 x 5 x \$58 + 2 x 5x \$58)	\$1,450
Step 4 (Prototype) Done in-house by the research team (year 3)	\$0
Step 5 (Test) Three focus groups with 4 transit users and two focus groups with 4 transit providers (year 3) (3 x 4 x \$58 + 2 x 4 x \$58)	\$1,160
Other: Codesign (Development of tools)	
Step 1 (Empathy) Four focus groups with 5 stakeholders (Year 4) (4 x 5 x \$58)	\$1,160
Steps 2 and 3 (Define and Ideate) Three focus groups with 5 transit users and 2 focus groups with 5 transit providers (Year 4) (3 x 5 x \$58 + 2 x 5x \$58)	\$1,450
Step 4 (Prototype) Done in-house by the research team (year 5)	\$0
Step 5 (Test) Three focus groups with 4 transit users and two focus groups with 4 transit providers (year 5) (3 x 4 x \$58 + 2 x 4 x \$58)	\$1,160
Total	\$7,540

Software (\$0)

This project will exclusively use open-source software.

Open access publishing (\$0)

We will leverage publishing agreements made by our institutions with reputable publishers such as Wiley, PLoS ONE, Elsevier, and Springer.



Funds from Other Sources

You must include all other sources of funding for the proposed research. Indicate whether these funds have been confirmed or not.

Full organization name Contribution type	Confirmed	Year 1 Year 5	Year 2	Year 3	Year 4
	☐				
	☐				
	☐				
	☐				
	☐				
	☐				
	☐				
	☐				
	☐				
Total funds from other sources		0	0	0	0
		0			

Personal information will be stored in the Personal Information Bank for the appropriate program.

'Application WEB'





Family name, Given name

Páez, Antonio

Suggested Reviewers

List Canadian or foreign specialists whom SSHRC may ask to assess your proposal.

List keywords that best describe the assessor's areas of research expertise. Please refer to the Suggested Assessors section of the detailed instructions for more information on conflicts of interest.

Family name Miller		Given name Harvey		Initials J	Title Professor
Org. code	Full organization name The Ohio State University		Keywords Transportation geography; accessibility; equity; GIScience; sustainable transportation		
Department/Division name Department of Geography			Address 1176 Derby Hall 154 North Oval Mall		
Country code 1		Area code 614	Number 29252907	City/Municipality Columbus	Prov./State OH
Telephone number		Extension		Postal/Zip code 43210	
Fax number		Country United States			
E-mail miller.81@osu.edu					
Family name Lovelace		Given name Robin		Initials	Title Professor
Org. code	Full organization name University of Leeds		Keywords transport modelling; public and active transport; open source software; open science; geocomputation		
Department/Division name Institute for Transport Studies			Address Room 2.04 Institute for Transport Studies 34-40 University Road		
Country code 44		Area code 0	Number 1133436496	Country United Kingdom	
Telephone number		Extension		Postal/Zip code	
Fax number		Country			
E-mail R.Lovelace@leeds.ac.uk					
Family name Golub		Given name Aaron		Initials	Title Professor
Org. code	Full organization name Portland State University		Keywords urban transportation planning; public transportation; policy; environmental justice; civil rights; sustainability		
Department/Division name College of Urban and Public Affairs			Address URBN 370E Portland State University 1825 SW Broadway		
Country code 1		Area code 503	Number 7254069	City/Municipality Portland	Prov./State OR
Telephone number		Extension		Postal/Zip code 97201	
Fax number		Country United States			
E-mail agolub@pdx.edu					

Personal information will be stored in the Personal Information Bank for the appropriate program.

'Application WEB'

Research Contributions: Antonio Páez

Refereed contributions BOOKS 2020-2026 (*: SSHRC supported; +: Student author)

1. Páez, A. & Boisjoly, G. (2026). *Discrete Choice Analysis with R*. Springer Cham.
<https://doi.org/10.1007/978-3-031-20719-8>

Refereed contributions JOURNAL ARTICLES 2020-2026 (*: SSHRC supported; +: Student author) (54 peer-reviewed journal publications in this period, only 30 most relevant to proposal are listed below)

- * 1. dos Santos, B. D. (+), Moghadasi, M. (+), & Páez, A. (2026). A 30-year analysis of active travel behaviour in Canada. *Journal of Transport Geography*, 131, 104561.
<https://doi.org/10.1016/j.jtrangeo.2026.104561>
- * 2. Higgins, C. D., Li, C., & Páez, A. (2026). Person-based aggregate space-time accessibility (PASTA): Bridging the gap between place- and person-based accessibility. *Journal of Transport Geography*, 132, 104602.
<https://doi.org/10.1016/j.jtrangeo.2026.104602>
- * 3. Soukhov, A. (+) Pereira, R. H. M., Higgins, C. D., & Páez, A. (2025). A family of accessibility measures derived from spatial interaction principles. *PLOS ONE*, 20(11), 1–31. <https://doi.org/10.1371/journal.pone.0335951>
- * 4. dos Santos, B. D. (+), Moghadasi, M. (+), & Páez, A. (2025). ActiveCA: Time use data from the general social survey of Canada to study active travel. *Environment and Planning B: Urban Analytics and City Science*, 52(8), 2037–2047.
<https://doi.org/10.1177/23998083251374724>
- * 5. Soukhov, A., Ravensbergen, L., Dorantes, L. M., & Páez, A. (2025). Towards completely caring 15-minute neighbourhoods. *Transportation Research Part A: Policy and Practice*, 198, 104503. <https://doi.org/10.1016/j.tra.2025.104503>
6. Elizondo-Candanedo, R. F. (+), Arranz-López, A., Soria-Lara, J. A., & Páez, A. (2024). When e-activities meet spatial accessibility: A theoretical framework and empirical space-time thresholds for simulated spatial settings. *Journal of Transport Geography*, 121, 104026.
<https://doi.org/10.1016/j.jtrangeo.2024.104026>
- * 7. Tavakoli, Z., Waygood, O., Abdollahi, S., & Páez, A. (2024). “Where Do Children Go?”: Exploring Children’s Daily Destinations with Children, Parents, and Experts. *Urban Planning*, 9(0). <https://doi.org/10.17645/up.8478>
- * 8. Tavakoli, Z. (+), Abdollahi, S. (+), Waygood, E. O. D., Páez, A., & Boisjoly, G. (2024). Traffic danger’s potential impact on children’s accessibility. *Transportation Research Part D: Transport and Environment*, 135, 104370. <https://doi.org/10.1016/j.trd.2024.104370>
9. Jamal, S. (+), & Páez, A. (2024). Well-being implications of immobility during COVID-19: Evidence from a student sample in Bangladesh using the satisfaction with life scale. *Transportation*, 51(5), 2019–2049. <https://doi.org/10.1007/s11116-023-10395-z>
- * 10. Desjardins, E., Lam, J., Reynard, D., Collins, D., Waygood, E.O.D., & Páez, A. (2024). Framing active school travel in Ontario, or how spinach is good for you. *Transportation Research Part A: Policy and Practice*, 180, 103953.
<https://doi.org/10.1016/j.tra.2024.103953>
- * 11. Soukhov, A. (+), Tarriño-Ortiz, J. (+), Soria-Lara, J. A., & Páez, A. (2024). Multimodal spatial availability: A singly-constrained measure of accessibility considering multiple modes. *PLOS ONE*, 19(2), e0299077. <https://doi.org/10.1371/journal.pone.0299077>
12. Rey-Blanco, D., Arbués, P., López, F. A., & Páez, A. (2024). Using machine learning to identify spatial market segments. A reproducible study of major Spanish markets. *Environment and Planning B: Urban Analytics and City Science*, 51(1), 89–108.
<https://doi.org/10.1177/23998083231166952>
- * 13. Merrall, J. (+), Higgins, C. D., & Páez, A. (2024). What’s a school worth to a

neighborhood? A spatial hedonic analysis of property prices in the context of accommodation reviews in Ontario. *Geographical Analysis*, 56(2), 217–243.
<https://doi.org/https://doi.org/10.1111/gean.12377>

- * 14. Soukhov, A. (+), **Páez, A.**, Higgins, C. D., & Mohamed, M. (2023). Introducing spatial availability, a singly-constrained measure of competitive accessibility. *PLOS ONE*, 18(1), e0278468. <https://doi.org/10.1371/journal.pone.0278468>
- * 15. Soukhov, A. (+), & **Páez, A.** (2023). TTS2016R: A data set to study population and employment patterns from the 2016 Transportation Tomorrow Survey in the Greater Golden Horseshoe Area, Ontario, Canada. *Environment and Planning B: Urban Analytics and City Science*, 50(2), 556–563. <https://doi.org/10.1177/23998083221146781>
- * 16. Sears, S. (+), Moataz, M., Ferguson, M., Razavi, S., & **Páez, A.** (2022). Perceived barriers to the movement of goods in Canada: A grounded theory investigation. *Transportation Research Part A: Policy and Practice*, 162, 27–45.
<https://doi.org/10.1016/j.tra.2022.05.011>
- * 17. Desjardins, E. (+), Tavakoli, Z. (+), **Páez, A.**, & Waygood, E. O. (2022). Children's access to non-school destinations by active or independent travel: A scoping review. In *International Journal of Environmental Research and Public Health*, No. 19; Vol. 19.
<https://doi.org/10.3390/ijerph191912345>
18. Desjardins, E. (+), Higgins, C. D., Scott, D. M., Apatu, E., & **Páez, A.** (2022). Correlates of bicycling trip flows in Hamilton, Ontario: Fastest, quietest, or balanced routes? *Transportation*, 49(3), 867–895. <https://doi.org/10.1007/s11116-021-10197-1>
19. Desjardins, E. (+), Higgins, C. D., & **Páez, A.** (2022). Examining equity in accessibility to bike share: A balanced floating catchment area approach. *Transportation Research Part D: Transport and Environment*, 102, 103091.
<https://doi.org/https://doi.org/10.1016/j.trd.2021.103091>
- * 20. Demitry, M. (+), Higgins, C. D., **Páez, A.**, & Miller, E. J. (2022). Accessibility to primary care physicians: Comparing floating catchments with a utility-based approach. *Journal of Transport Geography*, 101, 103356.
<https://doi.org/10.1016/j.jtrangeo.2022.103356>
- * 21. **Páez, A.**, & Higgins, C. D. (2021). An examination of the accessibility implications of a pilot COVID-19 vaccination program in Hamilton, Ontario. Findings.
<https://doi.org/10.32866/001c.24082>
22. **Páez, A.** (2021). Open spatial sciences: An introduction. *Journal of Geographical Systems*, 23(4), 467–476. <https://doi.org/10.1007/s10109-021-00364-4>
23. Pereira, R. H. M., Braga, C. K. V., Servo, L. M., Serra, B., Amaral, P., Gouveia, N., & **Páez, A.** (2021). Geographic access to COVID-19 healthcare in Brazil using a balanced float catchment area approach. *Social Science & Medicine*, 273.
<https://doi.org/10.1016/j.socscimed.2021.113773>
- * 24. Higgins, C. D., **Páez, A.**, Kim, G. (+), & Wang, J. (2021). Changes in accessibility to emergency and community food services during COVID-19 and implications for low income populations in Hamilton, Ontario. *Social Science & Medicine*, 291.
<https://doi.org/10.1016/j.socscimed.2021.114442>
25. Eldeeb, G. (+), Mohamed, M., & **Páez, A.** (2021). Built for active travel? Investigating the contextual effects of the built environment on transportation mode choice. *Journal of Transport Geography*, 96. <https://doi.org/10.1016/j.jtrangeo.2021.103158>
26. Desjardins, E. (+), Higgins, C. D., Scott, D. M., Apatu, E., & **Páez, A.** (2021). Using environmental audits and photo-journeys to compare objective attributes and bicyclists' perceptions of bicycle routes. *Journal of Transport & Health*, 22.
<https://doi.org/10.1016/j.jth.2021.101092>

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27. Desjardins, E. (+), Apatu, E., Razavi, S. D., Higgins, C. D., Scott, D. M., & **Páez, A.** (2021). “Going through a little bit of growing pains”: A qualitative study of the factors that influence the route choice of regular bicyclists in a developing cycling city. *Transportation Research Part F-Traffic Psychology and Behaviour*, 81, 431–444. <https://doi.org/10.1016/j.trf.2021.06.005>
28. Sarlas, G. (+), **Páez, A.**, & Axhausen, K. W. (2020). Betweenness-accessibility: Estimating impacts of accessibility on networks. *Journal of Transport Geography*, 84. <https://doi.org/10.1016/j.jtrangeo.2020.102680>
29. Jamal, S. (+), & **Páez, A.** (2020). Changes in trip-making frequency by mode during the COVID-19 emergency in Bangladesh. *Findings*. <https://doi.org/10.32866/001c.17977>
30. Jamal, S. (+), Mohiuddin, H., & **Páez, A.** (2020). How do the perceptions of neighborhood conditions impact active transportation? A study in Rajshahi, Bangladesh. *Transportation Research Part D-Transport and Environment*, 87. <https://doi.org/10.1016/j.trd.2020.102525>

Forthcoming contributions JOURNAL ARTICLES (*: SSHRC supported; +: Student author)

1. Soukhov, A. (+), Tiznado-Aitken, I., Palm, M., Farber, S., & **Paez, A.** (2025). Standards of fairness in the transportation literature. *Transportation*. <https://doi.org/10.1007/s11116-025-10693-8> (in press)
2. Soukhov, A. (+), Higgins, C. D., Mohamed, M., & **Páez, A.** (2025). Ten years of school closures and consolidations in Hamilton, Canada and the impact on multimodal accessibility. *Spatial Networks and Economics*. <https://doi.org/10.1007/s11067-025-09677-z> (in press)
3. Dos Santos, B.D. (+), Hansen, T., & **Páez, A.** (2026). Transportation and dignity: A scoping review and theoretical framework. *Transport Reviews* (revised and submitted)
4. Jamal, S., & **Páez, A.** (2026). Changes in walking frequency and its equity implications during the first wave of COVID-19 in Bangladesh. *Journal of Transport & Health* (revised and submitted)
5. Dos Santos, B.D. (+), Soukhov, A., & **Páez, A.** (2026) Equity Standards for transport policy: using objective accessibility, perceived accessibility and employment outcomes to set thresholds for equity planning. *Transportation Research Part A: Policy and Practice* (submitted)

Other research contributions (2020-2026)

- Between 2020-2026 I co-authored two reports (for Mobilizing Justice) and two book chapters.
- Editor-in-Chief, *Journal of Geographical Systems*.
- Editorial Board Member (*Annals of the Association of American Geographers*; *Economic and Business Letters*; *Geographical Analysis*; *International Journal of Geographical Information Science*; *Papers in Regional Science*; *Transportation*).
- 36 invited talks (17: Canada, 6: Spain, 4: Brazil, 3: USA, 2: Australia, 2: UK, 1: Italy, 1: Portugal).

Most significant career research contributions

A total of 177 publications in my record are indexed in Scopus. These publications have been cited 8,473 times (as of September 23, 2026). The average number of citations per indexed publication is 47.87, and the h-index is 50 (50 publications have been cited 50 times or more). Most significant career contributions are:

1. **Páez A.**, Scott DM, Morency C (2012). Measuring accessibility: positive and normative implementations of various accessibility indicators, *Journal of Transport Geography*, 25 141-153. 141–153. <https://doi.org/10.1016/j.jtrangeo.2012.03.016> (**656 citations**). This research was key to distinguishing between behavioral processes and policy aims in accessibility analysis. A key finding was the

identification of critical issues in the way accessibility measures are interpreted depending on their implementation. By providing concrete guidance for interpretation, this paper has become a key reference in accessibility research and has been cited in the fields of transportation, geography, economics, environmental studies, civil engineering, urban studies, and operations research.

2. **Páez, A., & Whalen, K. (+)** (2010). Enjoyment of commute: A comparison of different transportation modes. *Transportation Research Part A-Policy and Practice*, 44(7), 537–549. <https://doi.org/10.1016/j.tra.2010.04.003> (**273 citations**). This paper provided evidence of the intrinsic utility of active travel and challenged the conventional wisdom that travel time is always a disutility, with important implications for policy. Kate Whale was a Master student when this paper was written. It has become a valued reference for research in the fields of transportation, geography, business economics, public administration, and engineering.
3. **Páez A, Mercado RG (+), Farber S. (+), Morency C, Roorda M** (2010). Relative Accessibility Deprivation Indicators for Urban Settings: Definitions and Application to Food Deserts in Montreal, *Urban Studies*, 47 (7) 1415-1438. <https://doi.org/10.1177/0042098009353626> (**234 citations**). The methods presented in this paper were an early attempt at obtaining benchmarks for equity analysis. They provided the tools to assess differences in accessibility from a selected baseline, similar to the way poverty lines are measured with respect to some judiciously selected “median household”. This paper has become an important resource for research in geography, transportation, economics, environmental studies, public health, and urban studies.
4. Reyes, M.(+), **Páez, A., & Morency, C.** (2014). Walking accessibility to urban parks by children: A case study of Montreal. *Landscape and Urban Planning*, 125, 38–47. <https://doi.org/10.1016/j.landurbplan.2014.02.002> (**196 citations**). This research was led by Mario Reyes as a Master student, and it was among the first papers to examine the provision of green spaces from the perspective of children. This paper has been cited in the fields of environmental sciences, economics, public health, geography, and public administration.
5. Black J, **Páez A, Suthanaya P, (+)** (2002), Sustainable Urban Transportation: Performance Indicators and Some Analytical Approaches, *ASCE Journal of Urban Planning and Development*, 128 (4) 184-209 (**195 citations**). Sustainability has long been an interest of mine, particularly the social pillar of sustainability. This collaboration with John Black and Putu Suthanaya has been influential in civil engineering, transportation, urban studies, economics, and environmental science.

Contributions to training

Since 2020, I have supervised 8 senior undergraduate students in the programs of Engineering and Society (3), Geography (2), Life Sciences (2), and Economics (1). All these students completed a capstone project or thesis. I supervised 4 master’s students (3 in Geography, 1 in Public Health) and cosupervised 3 master students (2 in Geography and 1 in eHealth). I supervised 2 doctoral students (both in Geography, including a recipient of a Canada Graduate Scholarship award) and cosupervised 1 doctoral student in Civil Engineering. I currently supervise two master’s students in Geography and two doctoral students in Geography (including 1 Vanier Scholar). One master student (Desjardins) was the winner of the Master’s Thesis Competition in Transport Geography by the American Association of Geographers (2021). One doctoral student (Soukhov) was the winner of the Doctoral Thesis Competition in Transport Geography by the Association of American Geographers (2025). Graduate students working under my supervision are always offered opportunities to participate in research projects. Graduate students have participated in research leading to 37 out of a total of 54 peer-reviewed publications on record in the period since 2020, and their contributions are recognized with coauthorships.

Special circumstances

Since starting at McMaster in 2002, I have had four episodes of major depression. The first three were in 2005, 2013, and 2015. Unwisely, instead of taking the time necessary to heal at any of those opportunities, I pushed ahead with a work schedule that distracted me, but did not address my health needs. In 2025 I had another episode of major depression that was darker and longer than any of the preceding ones. Finally, I realized that I needed to recover, and was on medical leave from the university from November, 2025, until April 2026. This period of time away from work accounts for a gap in my teaching record and a decrease in my research activities, including grant writing and publishing.